

Day 53 - WAN Architectures

Wednesday, September 9, 2026

8:37 PM

WAN Info

Wide Area Network (WAN) is a network that extends over a large geographic area. It usually refers to an enterprise's private connections that connect their offices, data centers, and other sites together, it could also be used to refer to the Internet sometimes.

Over public/shared networks like the Internet, Virtual Private Networks (VPNs) can be used to create *private* WAN connections.

Types of WAN Connections

There are multiple types of WAN connection types, they include

1. **WAN over dedicated connection (Leased Line)**
2. **WAN over Ethernet (Fiber)**
3. **WAN over shared infrastructure (Internet VPN)**

1. Leased Lines

A *leased line* is a dedicated physical link, typically between two sites. They use serial connections (with *PPP* or *HDLC* encapsulation).

- Due to higher costs, higher installation times, and slower speeds of leased lines, Ethernet **WAN** technologies are becoming more popular.

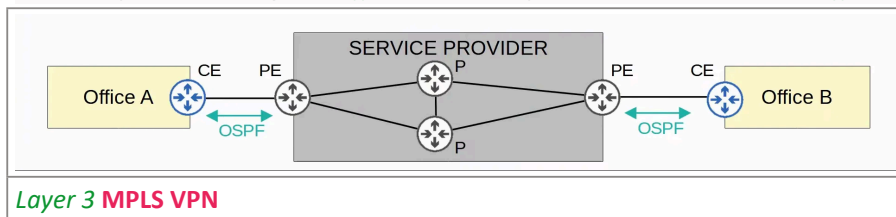
2. Multi Protocol Label Switching (MPLS)

Multi Protocol Label Switching (MPLS) is a shared infrastructure that lets many customer enterprises connect to a service provider's MPLS network and share the same infrastructure to make WAN connections.

- However, the **label switching** in the name in the name of **MPLS** allows **VPNs** to be created over the **MPLS** infrastructure through the use of *labels*.
- The three main parts of an **MPLS** are:
 1. **Customer Edge** router (**CE** router)
 2. **Provider Edge** router (**PE** router)
 3. **Provider Core** router (**P** router)
- When the **PE** routers receive frames from the **CE** routers, they add a *label* to the frame.
 - The *label* is placed in between the *Layer 2* header and the *Layer 3* header, so it's sometimes called a *Layer 2.5* protocol.
 - These *labels* are used to make forwarding decisions within the service provider network, not the *destination* IP.
- Many different technologies can be used to connect to a service provider's **MPLS** network for **WAN** service.

Layer 3 MPLS

- When using a *layer 3 MPLS VPN*, the **CE** and **PE** routers peer using **OSPF**, for example, to share routing information.
- The **CE** routers do not use **MPLS**, it is only use by the **PE/P** routers.
- For example, in the diagram, *Office A's CE* will peer with the one **PE**, and *Office B's CE* will peer with the other **PE**



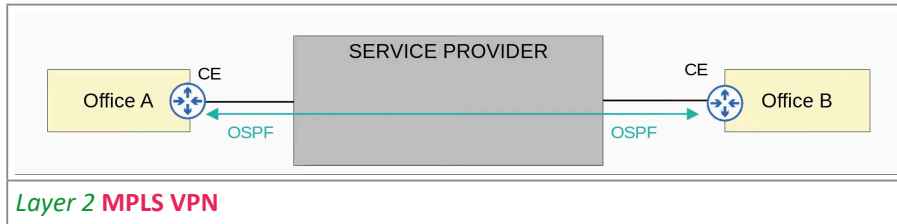
Layer 3 MPLS VPN

Layer 2 MPLS

- When using a *Layer 2 MPLS VPN*, the **CE** and **PE** routers do not form peerings.
- The service provider network is entirely transparent to the **CE** routers.
- In effect, it's like the **MPLS** network is a big switch and the two **CE** routers are directly connected.

- Their **WAN** interfaces will be in the same subnet.

- If a routing protocol is used (like **OSPF**), the two **CE** routers will peer directly with each other.



Internet Connections

There are countless ways for an enterprise to connect to the Internet.

- For example, the previously discussed private **WAN** technologies such as **leased lines** and **MPLS VPNs** can be used to connect to a service provider's Internet infrastructure
- In addition, technologies such as **CATV** and **DSL** commonly used by consumers for home Internet access, can also be used by an enterprise.
- These days, **Fiber Optic Ethernet** connections are growing in popularity for consumers and enterprises for their high speeds and long distances.

Digital Subscriber Line (DSL)

DSL provides Internet connectivity to customers over phone lines, and can share the same phone line that is already installed in most homes.

- A **DSL modem (modulator-demodulator)** is required to convert the data into a format suitable to be sent over the phone lines.
 - The **modem** can be a separate device or built into a 'home router'.

Cable TV Internet (CATV)

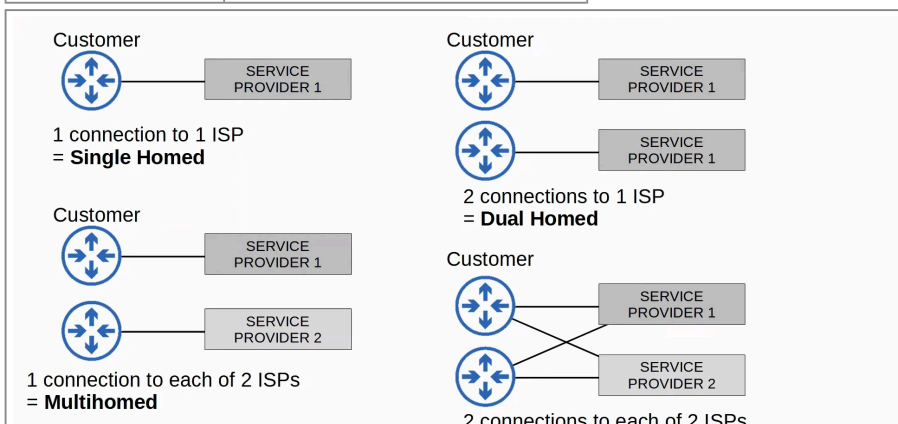
Similar to **DSL**, **CATV** Internet provides Internet access through the same **Cable Television (CATV)** lines used for TV service.

- A **cable modem** is required to convert the data into a format suitable to be sent over the **CATV** lines.
 - The **modem** can be a separate device or built into a 'home router'.

Redundant Internet Connections

For most enterprises, having a constant connection to the Internet is essential for their operations, that's why it's best for enterprises to set up redundant Internet connections, in order to avoid downtimes and heavy losses.

Term	Explanation
Single Homed	1 connection to 1 ISP
Dual Homed	2 connections to 1 ISP
Multihomed	1 connection to each of 2 ISPs
Dual Multihomed	2 connections to each of 2 ISPs



Internet VPNs

Private WAN services such as **leased lines** and MPLS provide security because each customer's traffic is separated by using dedicated physical connections (**leased line**) or by MPLS tags.

However, when using the Internet as a WAN to connect sites together, there is no built-in security features by default.

- In order to provide secure communications over the Internet, **VPNs** are used.
- There are two kinds of **VPNs** you need to know for the CCNA:
 - Site-to-Site VPNs using IPsec**
 - Remote-access VPNs using TLS**

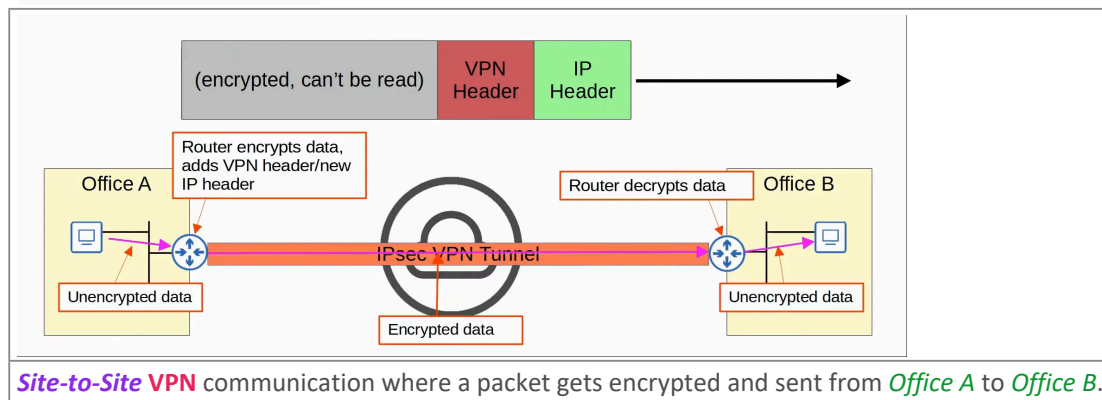
1. Site-to-Site VPNs

A '**site-to-site**' VPN is a VPN between two devices that is used to connect two sites together over the Internet.

- A **VPN** 'tunnel' is created between the two devices by encapsulating the original IP packet with a **VPN** header and a new IP header.
 - When **IPsec** is used, the original packet is encrypted before being encapsulated with the new header.
- In a **site-to-site VPN**, a tunnel is formed only between two tunnel endpoints (the two routers connected to the Internet in this example)
- All other devices in the site don't need to create a **VPN** tunnel for themselves. They can send unencrypted data to their site's router, which will do all the work of encrypting and forward it in the tunnel.

Process Summary

- The sending device combines the original packet and **session key (encryption key)** and runs them through an encryption formula.
- The sending device encapsulates the encrypted packet with a **VPN** header and a new IP header.
- The sending device sends the new packet to the device on the other side of the tunnel.
- The receiving device decrypts the data to get the original packet, and then forwards the original packet to its destination in the network.



Standard IPsec Limitations

- IPsec** doesn't support **broadcast** and **multicast** traffic, only **unicast**. This means that routing protocols such as **OSPF** can't be used over the tunnels, because they rely on **multicast** traffic. (A solution to this is [GRE over IPsec](#))
- Configuring a full mesh of tunnels between many sites is a labor-intensive task (A solution to this is [DMVPN](#))

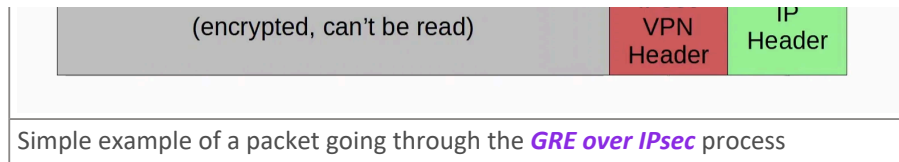
GRE over IPsec

Generic Routing Encapsulation (GRE) creates tunnels like IPsec, however it does not encrypt the original packet, so it is not secure.

However it has the advantage of being able to encapsulate a wide variety of **Layer 3** protocols as well as **broadcast** and **multicast** messages.

- To get the flexibility of **GRE** with the security of **IPsec**, we can use **GRE over IPsec**.
- The original packet will be encapsulated by a **GRE** header and a new IP header, and then the **GRE** packet will be encrypted and encapsulated within an **IPsec VPN** header and a new IP header.

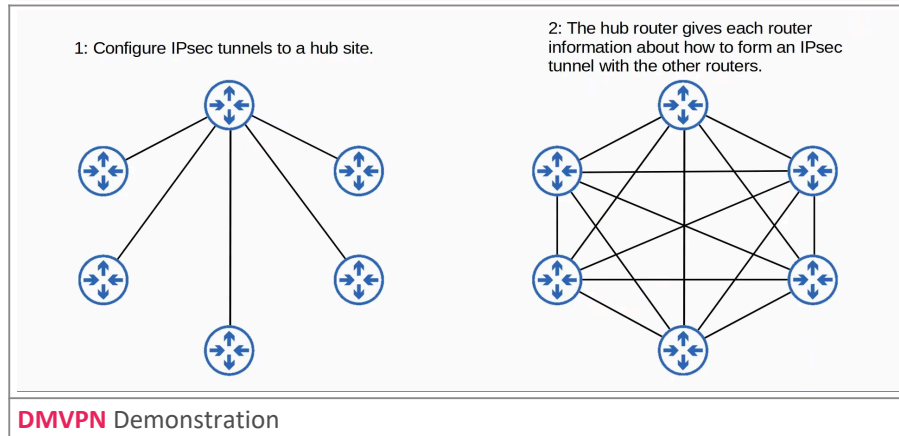




DMVPN

Dynamic Multipoint VPN (DMVPN) is a Cisco developed solution that allows routers to dynamically create a full mesh of IPsec tunnels without having to manually configure every single tunnel.

1. Configure **IPsec** tunnels to a **hub** site.
2. The **hub** router gives each router information about how to form an **IPsec** tunnel with the other routers.

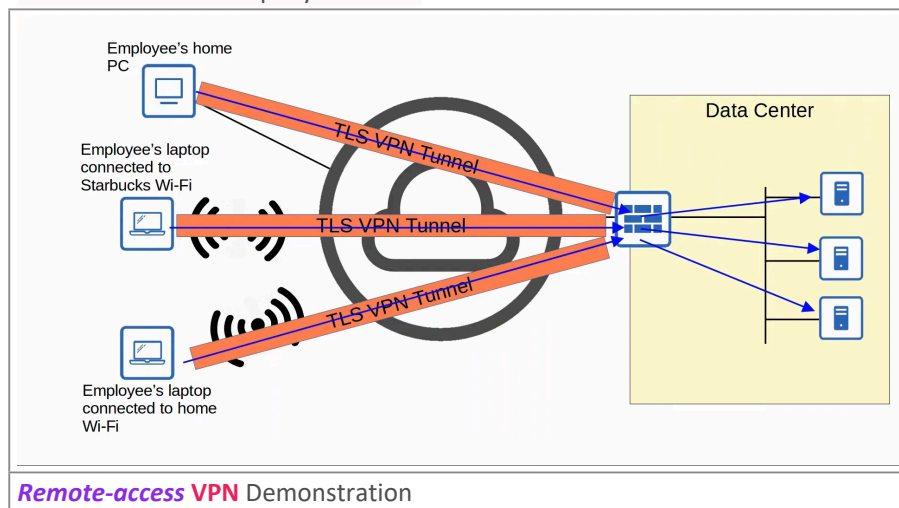


DMVPN Configuration

DMVPN provides the configuration simplicity of **hub-and-spoke** (each **spoke** router only needs one tunnel configured) and the efficiency of direct **spoke-to-spoke** communication (**spoke** routers can communicate directly without passing through the **hub**)

2. Remote-Access VPN

- Whereas **site-to-site VPNs** are used to make point-to-point connection between two sites over the Internet, **remote-access VPNs** are used to allow end devices (PCs, mobile phones) to access the company's internal resources securely over the Internet.
- Remote-access VPNs** typically use **Transport Layer Security (TLS)**.
 - TLS** is also what provides security for **HTTPS (HTTP Secure)**.
 - TLS** was formerly known as the **Secure Sockets Layer (SSL)**.
- VPN** client software is installed on end devices (for example to allow employees to connect to work from home). These end devices then form **secure** tunnels to one of the company's routers/firewalls acting as a **TLS** server.
- This allows the end users to securely access resources on the company's internal network without being directly connected to the company network.



Site-to-Site VS Remote-Access VPNs (Comparison)

1. Site-to-Site VPN

- They typically use **IPsec**.
- They provide service to many devices within the sites they are connecting.
- They are typically used to permanently connect two sites over the internet.

2. Remote-Access VPN

- They typically use **TLS**.
- They provide service to the one end device the **VPN** client software is installed on.
- They are typically used to provide on-demand access to end devices that want to securely access company resources while connected to a network which is not secure.